A-A-53289
30 November 1987

COMMERCIAL ITEM DESCRIPTION

VISCOMETER, Capillary Type

The General Services Administration has authorized the use of this commercial item description.

This Commercial Item Description covers one type of laboratory viscometer suitable for measurement of kinematic viscosity of transparent Newtonian liquids.

Salient Characteristics.

The Cannon-Fenske Routine capillary type viscometer shall be in strict accordance with applicable portions of American Society for Testing and Materials Designation (ASTM) D-446. The viscometer shall be suitable for testing in accordance with applicable portions of ASTM D-445.

Each capillary-type viscometer shall be calibrated in accordance with ASTM D-446 and shall be supplied with a calibration certificate and instructions for use. Calibration shall be by comparison with a master standard. The master standard shall be based on a viscosity of 1.0038 centistokes for water at 68 degrees Fahrenheit and shall be calibrated in accordance with ASTM designation 2162. Accuracy of master standard shall be traceable to National Bureau of Standards requirements. Accuracy of the viscometer shall be indicated by calibration of plus or minus 0.2 percent. Cannon-Fenske viscometers shall have the following constant, range and size designation as listed in TABLE I.

TABLE I. Cannon-Fenske Viscometers

ASTM Size	Approx. Constant (cSt/s)	Range (centistokes)
50	0.004	0.8-4
100	0.015	3.0-15
150	0.035	7.0-35
200	0.10	20.0-100
300	0.25	50.0-250
400	1.20	240.0-1200
450	2.50	500.0-2500

AMSC N/A FSC 6630

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The viscometer shall be legibly and permanently marked with the name or registered trademark of the manufacturer and the size.

Viscometer lot sizes shall be uniform in all characteristics and materials of construction for each size of viscometer. Viscometers shall have none of the following which shall be considered defects:

Missing components
Missing identification marking
Chips, cracks, protrusions
Striae, stones, bubbles, seeds

Workmanship. The viscometer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One viscometer of the size specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the items.

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Sampling for tests. Sampling for test shall be conducted in accordance with commercial practice to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more tests characteristics.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractor/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

For information only. The following products, as of the date of this document, are purported by the following companies to meet the salient characteristics of the item described herein:

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For lead NSN 6630-00-359-9903:

Beckman Instruments	Cat. No.	69-378-40
Cannon Instruments	Cat. No.	CFRC size 100
Fisher Scientific	Cat. No.	13-617C
Thomas Scientific	Cat. No.	9721-B59

NOTE: The following National Stock numbers are covered by this document.

Stock Number	Item Identification
6630-00-359-9903	VISCOMETER, CAPILLARY TYPE Size 100
6630-00-359-9904	VISCOMETER, CAPILLARY TYPE Size 200
6630-00-359-9905	VISCOMETER, CAPILLARY TYPE Size 300
6630-00-397-9293	VISCOMETER, CAPILLARY TYPE Size 50
6630-00-397-9305	VISCOMETER, CAPILLARY TYPE Size 150
6630-00-397-9309	VISCOMETER, CAPILLARY TYPE Size 400
6630-00-397-9311	VISCOMETER, CAPILLARY TYPE Size 450

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITY
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6630-0446

Location: NCR DISK #1

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INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53289	VISCOMETER, CAPILLARY TAPE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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